

# Climate Shocks and the Credit Market: A Monetary Policy Transmission Lens

Tichinashe Mabugu\*      Xolani Sibande†

August 14, 2026

AAAAAA

**Keywords:**

**JEL Codes:**

---

\*Corresponding Author. Economic Research Department, South African Reserve Bank; Email: xolani.sibande@resbank.co.za

†Economic Research Department, South African Reserve Bank; Email: tichinashe.mabugu@resbank.co.za.

# 1 Introduction

According to Bernanke and Gertler (1995), the credit channel is one of the key monetary policy transmission channels. Monetary policy affects the credit channel through its impact on the external finance premium <sup>1</sup>. Two linkages underscore monetary policy impacts on this premium: the balance sheet and the bank lending channel. The balance sheet channel stems from dynamics in the balance sheet and the incomes of borrowers, while the lending channel deals with how interest rates could impact lenders' loan supply volumes (Hall, 2001). The balance sheet channel has been well established as a significant monetary policy transmission channel (Aysun and Hepp, 2013; Altavilla et al., 2024; Ashcraft and Campello, 2007). Notwithstanding, there remains room to provide empirical estimations of the credit channel's strength due to recent developments in the identification of monetary policy shocks. This is especially true for emerging economies where this work is scant (Pirozhkova and Viegi, 2025). This paper hones in on the balance sheet channel by estimating how climate shocks can reduce (or indeed enhance) the effectiveness of unanticipated monetary policy decisions on the behaviour of borrowers.

The main motivation of this paper is that like monetary policy shocks, climate shocks – because of their impacts on capital and labour inputs and, in turn, their erosion of firm profitability – can cause frictions within the credit market and have consequential financial stability ramifications (Dafermos et al., 2018; Aglietta and Espagne, 2016). In so doing, climate shocks could widen the external finance premium and, in turn, stymie the efficacy of monetary policy pass-through. While this seems a likely course, there is also the potential for monetary policy to mute the deleterious climate shock effects on the credit market, in line with the literature that studies how monetary policy can be used as a tool to tackle climate change (Roy, 2024; Monasterolo and Raberto, 2018). In either case, climate shocks play a role in the functioning of the credit market and, consequently, the credit market monetary transmission channel.

---

<sup>1</sup>The external finance premium is governed by credit-market frictions which cause departures from lenders expected return and potential borrowers' expected costs.

I will investigate to what extent the manifestation of climate shocks act as a credit market friction and how this (potential) friction impacts monetary policy transmission. This involves three parts. First, I will compute the monetary policy shock variable. Second, I will estimate the causal relationships of interest in this work using a panel regression. Finally, I will use a quantile regression procedure to nuance these results.

Only physical shocks, not transition

## 2 Literature review

## 3 Conceptual Framework

We argue that climate shocks may alter the strength of the credit channel by affecting the financial position of corporates, households and banks. For simplicity, we illustrate the transmission of a contractionary monetary policy shock through the credit channel throughout this section. We distinguish between two possible effects that can ensue when a monetary policy shock coincides with a climate shock: amplification and dampening. We define amplification as a strengthening of the response of corporates or households to a monetary policy shock, such that the response of credit rates or volumes moves in the same direction as the response to the monetary policy shock alone, but with a larger absolute magnitude when the climate shock is present. A dampening effect, in contrast, occurs when the response continues to move in the same direction as the response to the monetary policy shock alone, but with a smaller absolute magnitude when the climate shock is present.

### 3.1 Corporates

From the perspective of corporates, we draw from a framework put forward by Bean et al (REF:Financial frictions and the monetary transmission mechanism: theory, evidence and policy implications 2002) as follows:

$$R_{it}^c = R_t + f(D_t/E_t) \quad f' > 0 \quad (1)$$

where  $R_{it}^c$  is the interest rate faced by corporates which is dependent on  $R_t$ , the risk-free interest rate and a function of the corporate's financial position, in this case represented by its Debt ( $D_t$ ) to Equity ( $E_t$ ) ratio. This function,  $f$  therefore represents the external finance premium. Insofar as there are information asymmetries or financial frictions, Equation 1 holds.

We posit that climate shocks alter the interest rate faced by corporate through changes in their financial position, which affect the external finance premium and, consequently, the cost of borrowing (REF: Laumer and Schaffer: Monetary policy transmission under supply chain pressure 2025). In particular, a climate shock can yield supply chain disruptions, damage of capital and production losses. These effects may reduce revenue and profits. Simultaneously, climate shock-related damage may reduce the value of collateral against which corporates borrow. These effects can increase corporate debt,  $D_t$ , while the associated decline in profitability, asset values and net worth can reduce equity,  $E_t$ . Consequently, a climate shock may increase a corporate's finance premium and in turn, raise the cost of external finance faced by the corporate. In this way, an adverse climate shock amplifies a contractionary monetary policy shock.

The same framework allows for a dampening effect of contractionary monetary policy. That is, firms with large cash buffers, liquid assets or adequate insurance coverage may experience little or no deterioration in their balance sheets, resulting in little or no change in  $D_t/E_t$  and consequently in the external finance premium,  $f(D_t/E_t)$ , following a climate shock. This resilience can also render corporates less responsive to the higher borrowing costs. Thus, climate shocks need not strengthen the transmission of monetary policy and may instead leave it unchanged or weaken it. Evidence that cash holdings can mitigate the effect of monetary tightening on corporate investment supports this mechanism (REF: Why Isn't Business Investment More Sensitive to Interest Rates? Evidence from Surveys. 2020. Steven A. Sharpe , Gustavo A. Suarez).

A final, related potential propagation mechanism for corporates arises from the possibility that corporate borrowing and investment decisions may be relatively less responsive to changes in borrowing costs. If climate shocks demand a need for adaptation, replacement of capital or reconstruction, corporates may have a strong incentive to undertake this investment even when

$R_t$  rises. In this case, corporates may become less sensitive to the cost of external finance. This does not necessarily mean that  $R_{it}^c$  itself falls; rather, the higher financing cost has a smaller effect on corporates' borrowing decisions. This effect would be most evident when higher interest rates coincide with an increase in credit volumes. Consistent with this possibility, Xu et al (REF: Navigating the green wave: Urban climate adaptation and firms' investment decisions-evidence from China 2025) demonstrate that adaptation policy increases firm investment even in the absence of an easing of financial constraints faced by firms.

### 3.2 Households

Analogous to Equation 1, we this of households facing an interest rate according to the following Equation:

$$R_{it}^h = R_t + \phi_t \quad (2)$$

where  $R_{it}^h$  is the interest rate faced by households as a result of the policy rate  $R_t$  and the household-specific spread between the policy rate and the cost of borrowing  $\phi_t$ . The spread can be viewed as reflecting household creditworthiness, indebtedness, income and collateral and other characteristics that affect lenders' assessment of credit risk and the terms at which credit is supplied. Thus,  $\phi_t$  allows households facing the same monetary policy rate to nevertheless face different borrowing costs. This allowance for differentiated credit market outcomes faced by households is driven by evidence suggesting that transmission of monetary policy varies across households according to a range of factors including their exposure to income fluctuations (REF:Household balance sheet channels of monetary policy: A back of the envelope calculation for the euro area BY Jiri Slacaleka, Oreste Tristani, Giovanni L. Violante).

The household mechanism ensues similarly to the case of firms. On the one hand, climate shocks may amplify the credit channel's response to monetary policy shocks. It follows that a climate shock may reduce household income, wealth or damage collateral. This deterioration in the balance sheet of households increases the household-specific spread,  $\phi_t$ , causing affected households to face higher borrowing costs for a given monetary policy rate. On the other hand, climate shocks may dampen this impact in a similar way to the case of corporates.

That is households with liquid assets, insurance coverage or other financial buffers may be able to absorb climate-related losses without a deterioration in their financial position or increased reliance on external finance. In such cases, the climate shock may generate little change in  $\phi_t$  and therefore need not amplify the response of household borrowing costs to monetary policy. Empirical evidence suggests that insurance may lower debt levels in the face of disasters as households use insurance payouts to pay off mortgages rather than to reconstruct (REF: Household Finance after a Natural Disaster: The Case of Hurricane Katrina. Justin Gallagher and Daniel Hartley).

### 3.3 Banks

The household and firm mechanisms are connected through the financial sector. Climate shocks can affect not only borrowers' demand for credit but also banks' willingness to supply it. There is a possibility that during climate shock episodes, creditors heighten the importance they attach to corporates' and households' balance sheet strength. Indeed, there is evidence that mortgage arrears is higher in regions that have faced more damage from climate shocks relative to less damaged regions (Ref: We didn't start the fire: Effects of a natural disaster on consumers' financial distress) Recent evidence suggests that in response, banks pose stricter lending conditions to borrowers in disaster-prone regions (REF: Natural Disaster Risk and Residential Mortgage Lending Standards. 2022. Jun Duanmu, Yongjia Li, Meimei Lin, Salman Tahsin).

On the other hand, a climate shock may dampen the transmission of monetary policy if it reduces the demand for credit. A deterioration in economic activity following a climate shock or adequate buffers to climate shocks may weaken corporates' and households' demand for new borrowing, reducing banks' loan demand and limiting the extent to which the shock translates into tighter credit conditions. When loan demand is weaker, banks may have less scope or incentive to adjust lending rates in response to changes in the riskiness of their borrowers. In this case, the climate shock may generate little additional increase in the spread between lending rates and the policy rate, thereby limiting its potential to amplify the increase in

borrowing costs associated with a monetary policy tightening.

While we present an array of explanations for how climate shocks interact with monetary policy transmission, we ultimately focus on whether the overall responsiveness of credit conditions to monetary policy shocks changes in the presence of climate shocks, without attributing this change to a specific household, corporate or banking-sector mechanism.

## 4 Data and methodology

### 4.1 Data

The analysis draws on three main data sources: climate shocks, monetary policy shocks, and credit extension data. The sample covers South Africa over the period 2011–2025. All series are observed at a quarterly frequency, determined by the availability of the monetary policy shock data; higher-frequency series are aggregated to quarterly values by taking averages or sums where appropriate.

**Climate data:** Temperature and precipitation data are sourced from the ERA5 reanalysis dataset produced by the European Centre for Medium-Range Weather Forecasts<sup>2</sup>. ERA5 provides globally gridded, hourly atmospheric estimates at a 31 km horizontal resolution. Grid-level values are weighted by population and aggregated to monthly averages over South Africa’s land area, from which we construct climate shock variables that capture deviations from long-run historical norms in temperature and precipitation.

The climate shocks are calculated as follows:

$$W_t = \frac{T_t - \bar{T}_t}{\sigma(T_t)}, \quad (3)$$

where  $W_t$  is the temperature or precipitation shock at time  $t$  and  $T_t$  is the observed population-weighted climate variable. For both temperature and precipitation,  $\bar{T}_t$  is the calendar-month mean computed over the full sample period (e.g., the mean of all January values from 2011 to 2025), and  $\sigma(T_t)$  is the corresponding within-month standard deviation.

---

<sup>2</sup>see <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-single-levels?tab=overview>

The resulting shock series are shown in Figure A3 in the Appendix.

**Monetary policy data:** The monetary policy shock series is constructed using high-frequency policy surprises for South Africa from Pirozhkova and Vieg (2025), alongside the repurchase (repo) rate from the South African Reserve Bank (SARB). The high-frequency surprises serve as an instrument to purge the endogenous component of policy rate changes, yielding a shock series that satisfies the exogeneity requirement of the regression framework described in Section 4.2. For robustness, we consider both market-based shocks from Pirozhkova and Vieg (2025) (see Figure A4) and a theory-based shock derived from the policy rate, as described in Section 4.2 below (see Figure A5). Surprises are timed to coincide with policy rate announcement dates and aggregated to a quarterly frequency. Underlying the calculation of the are the forecasts of the real GDP growth, the policy rate (repo-rate), and the inflation rate from Bloomberg Consensus.

**Credit extension data:** Private sector credit extension data — disaggregated by borrower type (households and corporates) at the bank level — are obtained from the SARB’s BA900 and BA930 bank compliance reports.<sup>3</sup> These series include both credit volumes and lending rates, and are constructed following Sibande et al. (2025) and Sibande and Milne (2025) by sorting and aggregating individual bank records into the broad credit categories used in the study: unsecured, secured, and mortgage credit. Summary charts are provided in Figure A1 and Figure A2.

## 4.2 Methodology

**Structural shocks:** Rather than using the policy interest rate directly — which responds to prevailing economic conditions and is therefore endogenous — we use an unexpected change in monetary policy, commonly called a monetary policy shock, to isolate the causal effect of policy on credit markets (Miyajima, 2024). We construct this shock following Merrino (2022), who adapts the approach of Romer and Romer (2004) to the South African context. The idea is to regress the change in the repo rate forecast on variables that policymakers were already

---

<sup>3</sup>BA930 data at the bank level are not publicly available.

tracking, and treat whatever is left over — the part of the rate change that cannot be explained by those forecasts — as the true policy surprise. Formally:

$$\Delta r_t = \alpha + \tilde{r}_t + \sum_{t=0}^2 \beta_t \Delta \tilde{y}_t + \sum_{t=0}^2 \delta_t \tilde{\pi}_t + \varepsilon_t, \quad (4)$$

where  $\Delta r_t$  is the change in the repo rate forecast,  $\tilde{r}_t$  is the forecasted repo rate,  $\tilde{y}_t$  is the forecasted real GDP growth rate, and  $\tilde{\pi}_t$  is the forecasted inflation rate. The residual  $\varepsilon_t$  — the unexplained part of the forecast revision — is our monetary policy shock,  $\text{MPShock}_t$ .<sup>4</sup>

As an alternative, we construct a second measure of the monetary policy surprise following Miyajima (2024). This approach uses forecast errors — the gap between what forecasters expected and what actually occurred — rather than SARB forecasts. The construction proceeds in two steps. First, for each variable  $i$ , the forecast error is:

$$FE_t^i = ACT_t^i - EXP_t^i, \quad (5)$$

where  $i \in \{\text{policy rate, inflation, GDP growth}\}$ ,  $ACT_t^i$  is what was actually observed, and  $EXP_t^i$  is the Bloomberg consensus forecast made beforehand. In the second step, the policy rate forecast error is regressed on the inflation and output growth forecast errors:

$$FE_t^r = \alpha + \beta FE_t^\pi + \gamma FE_t^y + \epsilon_t, \quad (6)$$

where  $FE_t^\pi$  is the inflation forecast error and  $FE_t^y$  is the GDP growth forecast error. The residual  $\epsilon_t$  is the monetary policy surprise — the part of the policy rate miss that has nothing to do with inflation or growth news, and is therefore not anticipated by market participants.

**Market-based shocks:** As a further robustness check, we use market-based monetary policy surprises built from a high-frequency identification approach (Gürkaynak et al., 2005). The core idea is to measure how financial market prices move in a narrow window around each MPC announcement: since little else happens in that window, any price change can be attributed to the policy news itself (Nakamura and Steinsson, 2018). Following Pirozhkova

---

<sup>4</sup>We include forecasts up to two quarters ahead to account for the common assumption among forecasters that policy is unchanged in the very near term. See Romer and Romer (2004) for details.

and Viegi (2025), we decompose the overall surprise into four components: a target factor (the surprise in the level of the policy rate), a forward guidance factor (the surprise in the expected future path of policy), a central bank information factor (what the central bank’s communication revealed about the economic outlook that markets did not already know), and a country risk factor (shifts in sovereign risk perceptions around MPC dates). Each component is used in turn as an alternative measure of  $\text{MPShock}_t$  in Equation 8 and Equation 9.

**Shock purging:** Policymakers are aware of climate conditions and may already adjust rates in response to them. If so, our monetary policy shocks would partly reflect climate events rather than pure policy decisions, which would confound our estimates. To address this, we regress each monetary policy shock on the climate shock measures:

$$\text{MPShock}_t^i = \text{ClimateShock}_t^i + \varepsilon_t^i, \quad (7)$$

where  $\text{MPShock}_t^i$  is each of the monetary policy shocks described above and  $\text{ClimateShock}_t^i$  are the temperature and precipitation shocks. The residual  $\varepsilon_t^i$  — the part of the policy shock that cannot be explained by climate conditions — is retained as a “purged” shock. We use these purged shocks in all subsequent estimations.

**Panel estimation:** Our first estimating equation tests whether the effect of a monetary policy shock on credit is amplified or dampened when a climate shock occurs at the same time:

$$\begin{aligned} \text{Credit}_{s,t} = & \beta_0 + \beta_1 \text{Climate}_t^k + \beta_2 \text{MPShock}_t + \beta_3 (\text{Climate}_t^k \times \text{MPShock}_t) + \\ & \Gamma X_t + \gamma_s + \delta_t + \varepsilon_{s,t}, \end{aligned} \quad (8)$$

where  $\text{Credit}_{s,t}$  is credit growth for agent  $s$  (households or corporates) at time  $t$ , and  $\text{Climate}_t^k$  captures climate shocks — temperature, precipitation, and extreme wet and dry events, indexed by  $k$ . The interaction term  $\text{Climate}_t^k \times \text{MPShock}_t$  is the key coefficient of interest: a significant estimate would indicate that climate shocks alter how monetary policy transmits through credit markets.  $\gamma_s$  are agent fixed effects (absorbing time-invariant differences across borrower types),  $\delta_t$  are time fixed effects, and  $\varepsilon_{s,t}$  is the error term.  $X_t$  is a vector of controls added as a robustness check, including **GDP growth, the USD/ZAR exchange**

rate, the loan-to-deposit ratio, and the unemployment rate.

**Panel local projections:** The panel regression above captures the contemporaneous relationship but cannot tell us how long any effect lasts. To trace the full dynamic response of credit over time, we use the local projections method of Jordà (2005), estimating a separate regression for each future horizon  $h = 0, 1, \dots, H$ :

$$\begin{aligned} Credit_{s,t+h} - Credit_{s,t-1} = & \beta_0 + \beta_1 Climate_t^k + \beta_2 MPShock_t + \beta_3 (Climate_t^k \times MPShock_t) + \\ & \Gamma X_t + \gamma_s + \delta_t + \varepsilon_{s,t}, \end{aligned} \tag{9}$$

where  $Credit_{s,t+h} - Credit_{s,t-1}$  is the cumulative change in credit growth from period  $t-1$  to  $t+h$ , and  $h$  is the horizon in quarters. Plotting the estimated  $\hat{\beta}_1$ , and  $\hat{\beta}_2 + \hat{\beta}_3$  across horizons gives the impulse response functions, showing the differential impact of climate on credit in the quarters following a shock. **All regressions use Driscoll-Kraay standard errors to account for correlation across agents and over time (Driscoll and Kraay, 1998).**

## 5 Results

### 6 Drivers

#### 6.1 Timing sensitivity

#### 6.2 Sectoral sensitivity

#### 6.3 Type of shock

## 7 Conclusion

Table 1: Estimated coefficients from the fixed effects models

| term                                    | (1)       | (2)      | (3)       | (4)       | (5)       | (6)       | (7)       | (8)       | (9)       | (10)     | (11)      | (12)      |
|-----------------------------------------|-----------|----------|-----------|-----------|-----------|-----------|-----------|-----------|-----------|----------|-----------|-----------|
| pop_temp_shock                          | 1.062***  |          |           |           |           |           |           |           |           |          |           |           |
| myajima_surprise                        | -0.764*** |          |           |           |           |           |           |           |           |          |           |           |
| pop_temp_shockmyajima_surprise          | 1.361***  |          |           |           |           |           |           |           |           |          |           |           |
| pop_temp_shock                          |           | 1.025*** |           |           |           |           |           |           |           |          |           |           |
| romer_surprise                          |           | 0.107*   |           |           |           |           |           |           |           |          |           |           |
| pop_temp_shockromer_surprise            |           | 0.533*** |           |           |           |           |           |           |           |          |           |           |
| pop_temp_shock                          |           |          | 1.09***   |           |           |           |           |           |           |          |           |           |
| target                                  |           |          | -0.013*** |           |           |           |           |           |           |          |           |           |
| pop_temp_shocktarget                    |           |          | -0.016*** |           |           |           |           |           |           |          |           |           |
| pop_temp_shock                          |           |          |           | 1.172***  |           |           |           |           |           |          |           |           |
| forward_guidance                        |           |          |           | 0.024***  |           |           |           |           |           |          |           |           |
| pop_temp_shockforward_guidance          |           |          |           | -0.032*** |           |           |           |           |           |          |           |           |
| pop_temp_shock                          |           |          |           |           | 1.01***   |           |           |           |           |          |           |           |
| central_bank_information                |           |          |           |           | 0.063***  |           |           |           |           |          |           |           |
| pop_temp_shockcentral_bank_information  |           |          |           |           | -0.041*** |           |           |           |           |          |           |           |
| pop_temp_shock                          |           |          |           |           |           | 1.58***   |           |           |           |          |           |           |
| country_risk                            |           |          |           |           |           | -0.023*** |           |           |           |          |           |           |
| pop_temp_shockcountry_risk              |           |          |           |           |           | 0.16***   |           |           |           |          |           |           |
| pop_presp_shock                         |           |          |           |           |           |           | -0.078*** |           |           |          |           |           |
| myajima_surprise                        |           |          |           |           |           |           | 1.142***  |           |           |          |           |           |
| pop_presp_shockmyajima_surprise         |           |          |           |           |           |           | 0.413***  |           |           |          |           |           |
| pop_presp_shock                         |           |          |           |           |           |           |           | -0.054*** |           |          |           |           |
| romer_surprise                          |           |          |           |           |           |           |           | 0.551***  |           |          |           |           |
| pop_presp_shockromer_surprise           |           |          |           |           |           |           |           | -0.104*** |           |          |           |           |
| pop_presp_shock                         |           |          |           |           |           |           |           |           | -0.028*** |          |           |           |
| target                                  |           |          |           |           |           |           |           |           | -0.037*** |          |           |           |
| pop_presp_shocktarget                   |           |          |           |           |           |           |           |           | 0         |          |           |           |
| pop_presp_shock                         |           |          |           |           |           |           |           |           |           | -0.014** |           |           |
| forward_guidance                        |           |          |           |           |           |           |           |           |           | 0.034*** |           |           |
| pop_presp_shockforward_guidance         |           |          |           |           |           |           |           |           |           | 0.005*** |           |           |
| pop_presp_shock                         |           |          |           |           |           |           |           |           |           |          | -0.032*** |           |
| central_bank_information                |           |          |           |           |           |           |           |           |           |          | 0.044***  |           |
| pop_presp_shockcentral_bank_information |           |          |           |           |           |           |           |           |           |          | 0.004***  |           |
| pop_presp_shock                         |           |          |           |           |           |           |           |           |           |          |           | -0.063*** |
| country_risk                            |           |          |           |           |           |           |           |           |           |          |           | 0.018***  |
| pop_presp_shockcountry_risk             |           |          |           |           |           |           |           |           |           |          |           | 0.037***  |

## References

- Aglietta, M. and Espagne, É. (2016). *Climate and Finance Systemic Risks, More than an Analogy?: The Climate Fragility Hypothesis*. CEPII, Centre d’études prospectives et d’informations internationales.
- Altavilla, C., Gürkaynak, R. S., and Quaedvlieg, R. (2024). Macro and micro of external finance premium and monetary policy transmission. *Journal of Monetary Economics*, 147:103634.
- Ashcraft, A. B. and Campello, M. (2007). Firm balance sheets and monetary policy transmission. *Journal of Monetary Economics*, 54(6):1515–1528.
- Aysun, U. and Hepp, R. (2013). Identifying the balance sheet and the lending channels of monetary transmission: A loan-level analysis. *Journal of banking & Finance*, 37(8):2812–2822.
- Bernanke, B. S. and Gertler, M. (1995). Inside the black box: The credit channel of monetary policy transmission. *Journal of Economic perspectives*, 9(4):27–48.
- Dafermos, Y., Nikolaidi, M., and Galanis, G. (2018). Climate change, financial stability and monetary policy. *Ecological Economics*, 152:219–234.
- Driscoll, J. C. and Kraay, A. C. (1998). Consistent covariance matrix estimation with spatially dependent panel data. *Review of Economics and Statistics*, 80(4):549–560.
- Gürkaynak, R. S., Sack, B., and Swanson, E. (2005). Do actions speak louder than words? the response of asset prices to monetary policy actions and statements. *International Journal of Central Banking*, 1(1):55–93.
- Hall, S. (2001). Credit channel effects in the monetary transmission mechanism. *Bank of England. Quarterly Bulletin*, 41(4):442.
- Jordà, O. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1):161–182.

- Merrino, S. (2022). Monetary policy and wage inequality in South Africa. *Emerging Markets Review*, 53:100911.
- Miyajima, K. (2024). Monetary policy, inflation and distributional impact: South Africa's case. *South African Journal of Economics*, 92(3):305–330.
- Monasterolo, I. and Raberto, M. (2018). The EIRIN flow-of-funds behavioural model of green fiscal policies and green sovereign bonds. *Ecological Economics*, 144:228–243.
- Nakamura, E. and Steinsson, J. (2018). High-frequency identification of monetary non-neutrality: The information effect. *Quarterly Journal of Economics*, 133(3):1283–1330.
- Pirozhkova, E. and Vieg, N. (2025). The bank lending channel of monetary policy transmission in South Africa. *The Quarterly Review of Economics and Finance*, page 102041.
- Romer, C. D. and Romer, D. H. (2004). A new measure of monetary shocks: Derivation and implications. *American economic review*, 94(4):1055–1084.
- Roy, A. (2024). Green monetary policy to combat climate change: Theory and evidence of selective credit control. *Journal of Climate Finance*, 6:100035.
- Sibande, X. and Milne, A. (2025). The Impact of Basel III Implementation on Bank Lending in South Africa. *South African Journal of Economics*, 93(1):6–26.
- Sibande, X., Nxumalo, D., Mncube, K., Koch, S., and Vieg, N. (2025). Regulation and Bank Lending in South Africa: A Narrative Index Approach. *South African Journal of Economics*, 93(1):73–85.

# A Appendix

## A.1 Data

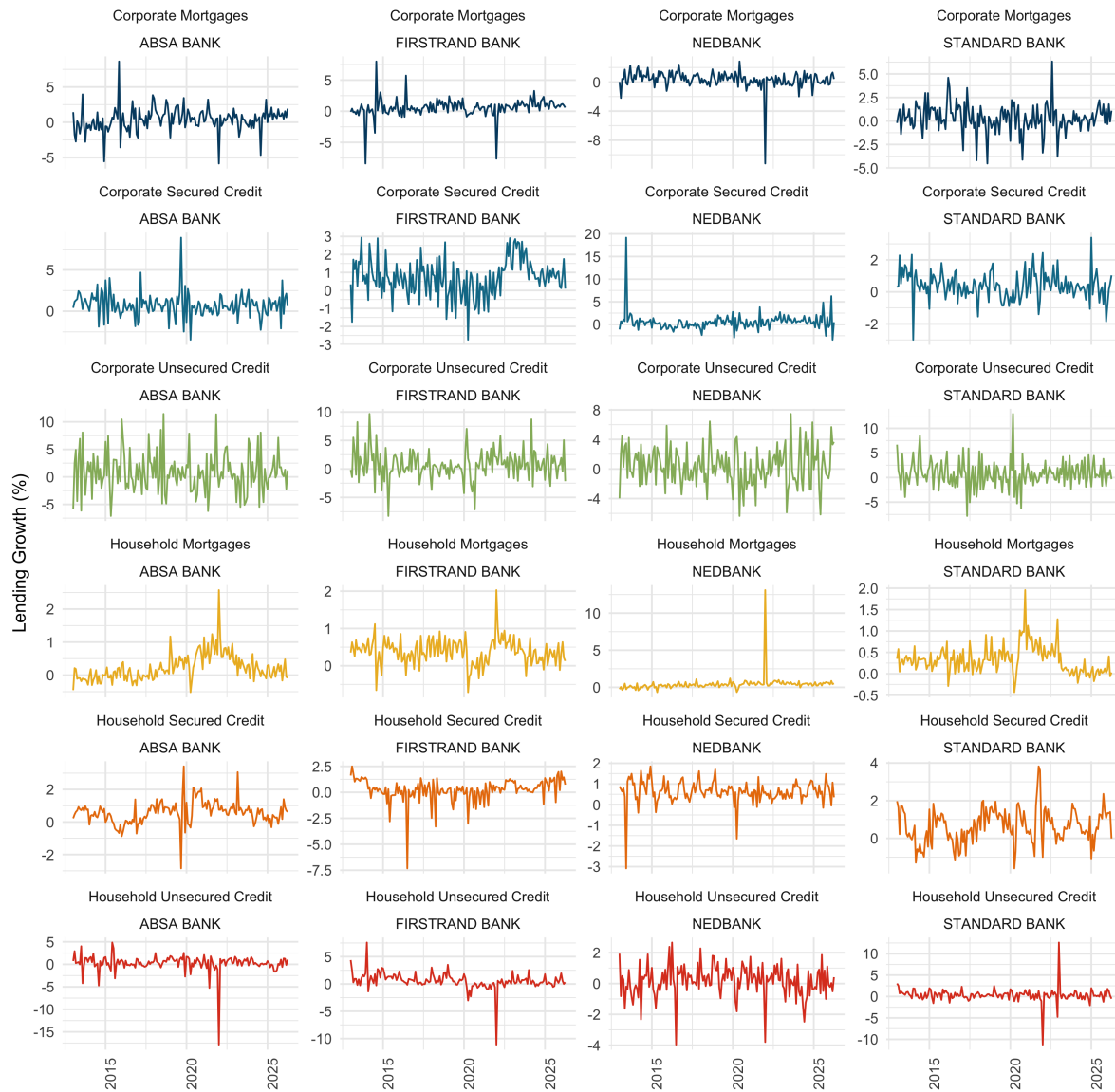


Figure A1: Lending volume growth over time

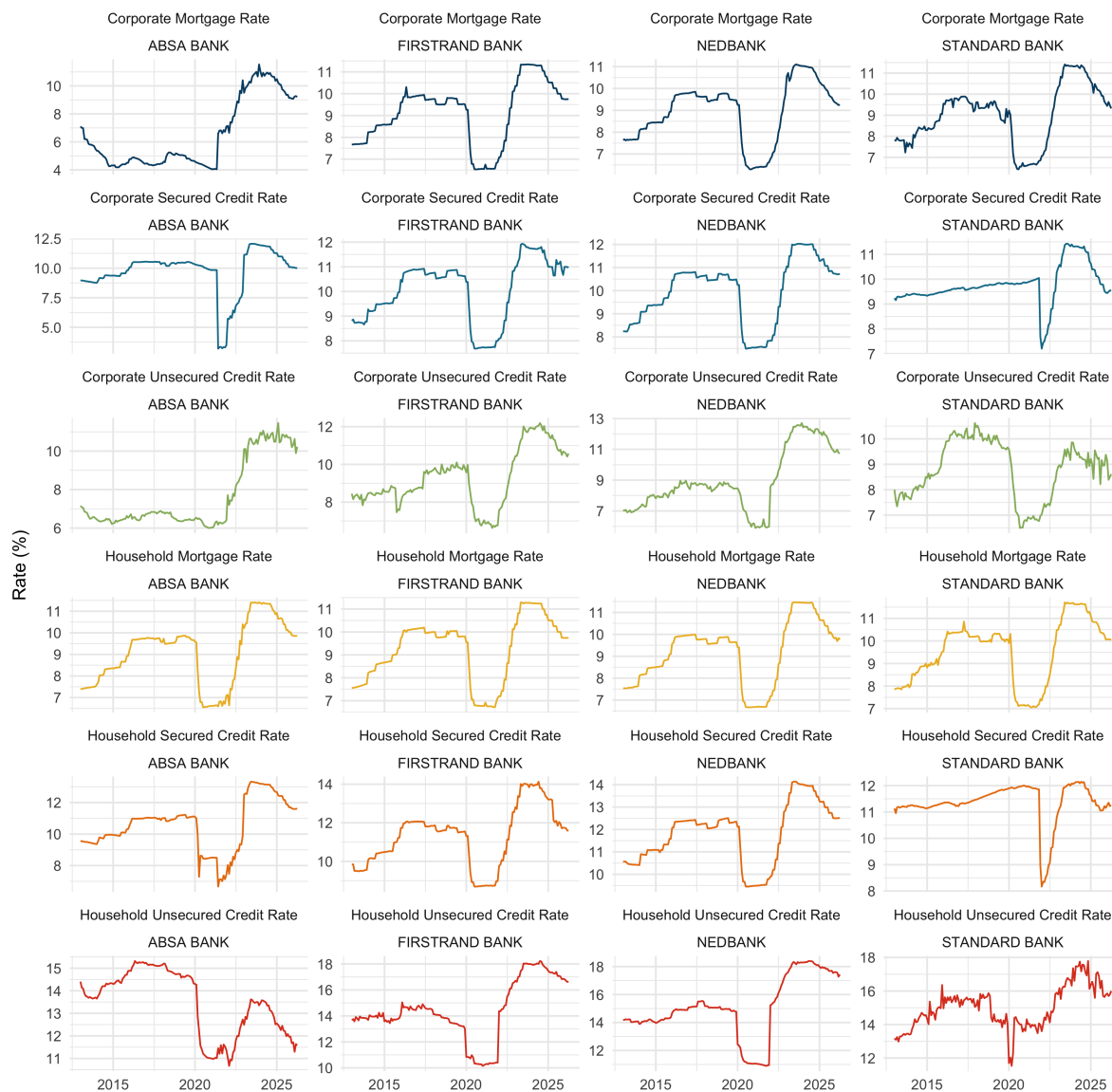


Figure A2: Lending rates overtime

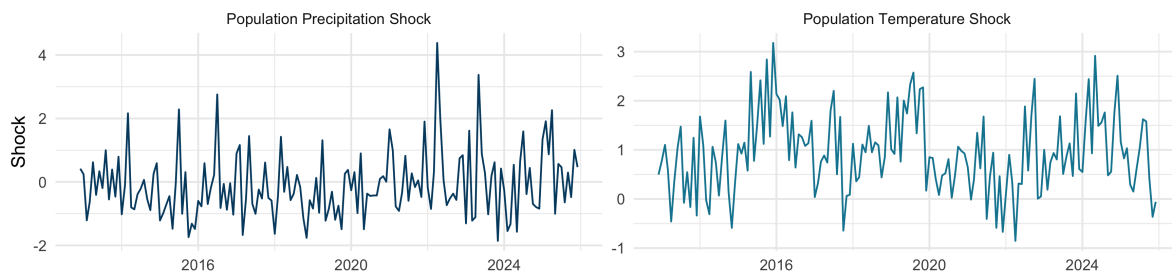


Figure A3: Climate Shocks Data

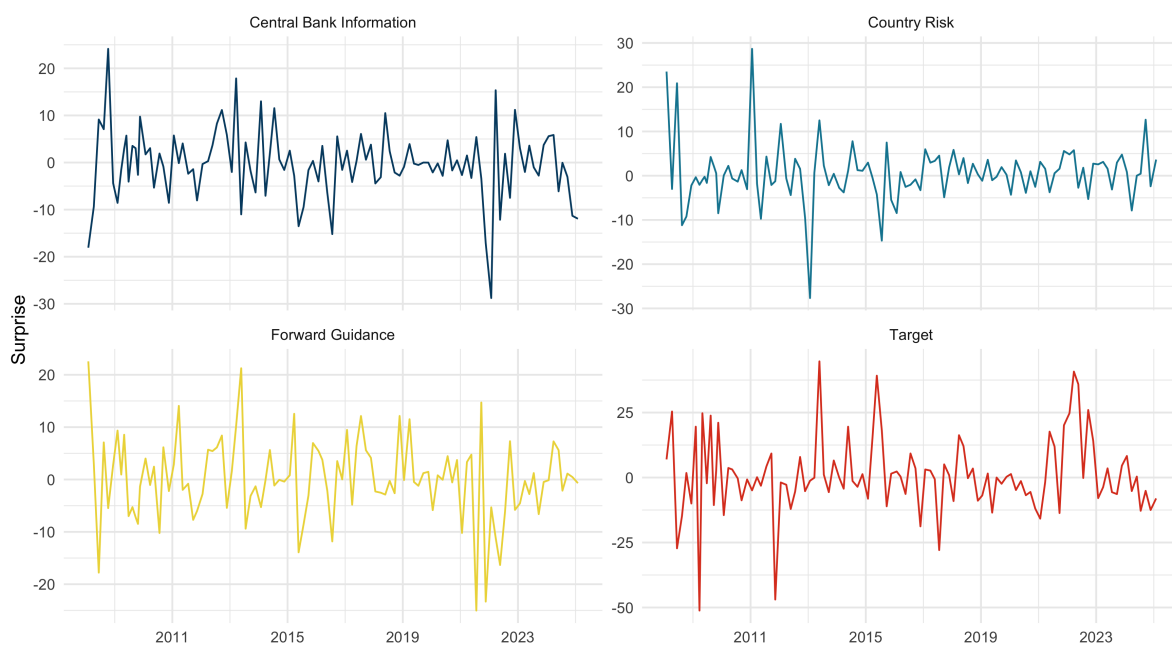


Figure A4: Market-Based Surprises Data

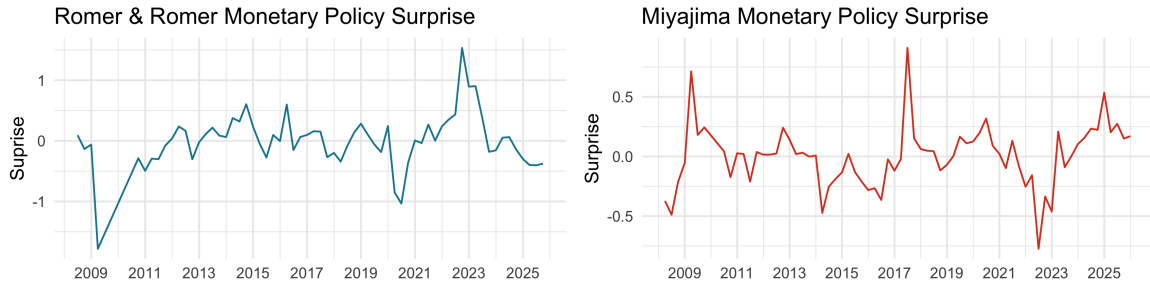


Figure A5: Theory-based surprises data

## A.2 Descriptive Statistics

Table A1: Descriptive statistics

| Variable         |                                    | Mean  | Median | SD    | Min    | Max   | N   |
|------------------|------------------------------------|-------|--------|-------|--------|-------|-----|
| Monetary Policy  |                                    |       |        |       |        |       |     |
|                  | Central Bank Information           | -1.64 | -0.42  | 5.69  | -17.14 | 11.19 | 192 |
|                  | Country Risk                       | 0.64  | 0.47   | 3.38  | -5.47  | 12.50 | 192 |
|                  | Forward Guidance                   | -0.25 | -0.47  | 6.93  | -23.32 | 21.24 | 192 |
|                  | Miyajima Surprise                  | -0.00 | 0.01   | 0.26  | -0.77  | 0.91  | 208 |
|                  | Romer Surprise                     | 0.06  | 0.06   | 0.41  | -1.04  | 1.53  | 204 |
|                  | Target                             | 3.12  | 0.64   | 13.46 | -18.78 | 44.70 | 192 |
| Corporate Credit |                                    |       |        |       |        |       |     |
|                  | Corporate Mortgage Rate            | 8.43  | 9.07   | 2.04  | 4.06   | 11.35 | 212 |
|                  | Corporate Mortgages                | 74.39 | 75.79  | 6.99  | 24.93  | 77.82 | 212 |
|                  | Corporate Secured Credit           | 71.57 | 72.62  | 6.62  | 24.16  | 74.80 | 212 |
|                  | Corporate Secured Credit Rate      | 9.91  | 9.99   | 1.36  | 3.30   | 12.06 | 212 |
|                  | Corporate Unsecured Credit         | 76.25 | 77.16  | 7.01  | 25.81  | 79.22 | 212 |
|                  | Corporate Unsecured Credit Rate    | 8.71  | 8.62   | 1.74  | 5.95   | 12.59 | 212 |
| Household        |                                    |       |        |       |        |       |     |
| Credit           |                                    |       |        |       |        |       |     |
|                  | Household Mortgage Rate            | 9.30  | 9.75   | 1.41  | 6.56   | 11.67 | 212 |
|                  | Household Mortgages                | 77.38 | 78.22  | 7.16  | 26.14  | 80.28 | 212 |
|                  | Household Secured Credit           | 73.94 | 75.28  | 6.87  | 24.99  | 77.44 | 212 |
|                  | Household Secured Credit Rate      | 11.28 | 11.35  | 1.42  | 7.09   | 14.11 | 212 |
|                  | Household Unsecured Credit         | 73.93 | 75.12  | 6.86  | 24.58  | 76.35 | 212 |
|                  | Household Unsecured Credit Rate    | 14.52 | 14.54  | 1.96  | 10.22  | 18.37 | 212 |
|                  | Population Precipitation Shock     | -0.26 | -0.48  | 1.88  | -4.54  | 6.38  | 204 |
|                  | Population Precipitation Shock Abs | 1.36  | 1.12   | 0.90  | 0.00   | 4.54  | 204 |
|                  | Neg                                |       |        |       |        |       |     |
|                  | Population Precipitation Shock Pos | 1.10  | 0.76   | 1.29  | 0.00   | 6.38  | 204 |
|                  | Population Temperature Shock       | 0.95  | 1.06   | 0.60  | -0.26  | 2.43  | 204 |
|                  | Population Temperature Shock Abs   | 0.04  | 0.00   | 0.09  | 0.00   | 0.42  | 204 |
|                  | Neg                                |       |        |       |        |       |     |
|                  | Population Temperature Shock Pos   | 0.99  | 1.06   | 0.55  | 0.05   | 2.43  | 204 |

### A.3 Sector Credit Exposure

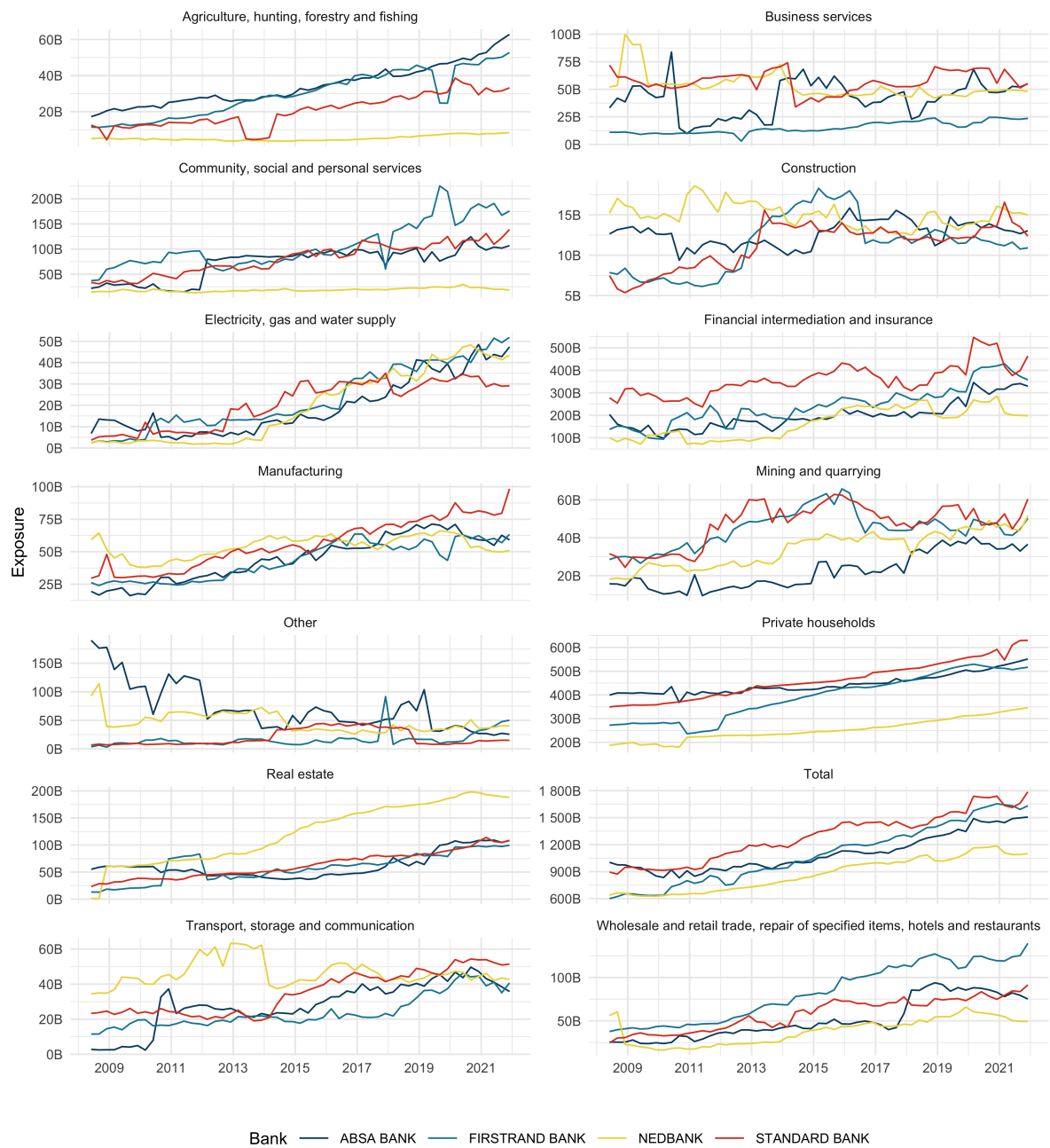


Figure A6: Sector credit exposure